

Can We Trust AI More After Cross-Checking?

COMP2501 evidence report

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This report is a presentation companion. The course handout requires an in-class presentation and proposal; it does not separately require this PDF. Public source rows and R scripts are linked below. The revised 500-question mathematics bank contains user-local material, so only its R-audited aggregate results are included.

What we measured

The outcome is a **wrong final answer**, not a failure to give any answer. For each named, scorable condition, $\text{wrong-answer rate} = \text{incorrect} / (\text{correct} + \text{incorrect} + \text{explicit abstention})$. A clear “I cannot determine this” response remains in the denominator but is **not** an error. Transport failures, incomplete outputs, and unscorable replies are outside that denominator and shown separately. This distinction does not certify that the reasoning behind a correct final answer is sound.

The study contains several **different** data collections. Do not add their rows or compare their rates as though the questions, model routes, prompts, or selection rules were the same. The R-generated [15-row evidence inventory](#) keeps the public studies separate and records counts, denominators and provenance. The revised 500-question MiniMax-only mathematics study uses a private bank with user-local materials; only its aggregate results appear below, not its raw questions or responses.

Related work and our narrower contribution

[Huang et al. \(ICLR 2024\)](#) found that models can struggle to correct reasoning without external feedback and sometimes worsen an answer. [Dhuliawala et al. \(ACL Findings 2024\)](#) reduced hallucinations with Chain-of-Verification, which asks and answers separate verification questions independently; our neutral follow-up is a simpler check, not a replication of that method. [Sharma et al. \(ICLR 2024\)](#) observed sycophancy across five assistants and four free-form tasks, motivating our scripted false-user-premise condition without predicting the error rate of the current models. [Mirzadeh et al. \(2024\)](#) documented brittleness of mathematical word-problem performance under controlled variations; our selected mathematics questions are a separate empirical test. The papers motivate competing mechanisms and designs. They do not substitute for the paired rates below.

Results aligned to the four research questions

1. Can double-checking reduce AI error rates?

Yes, in the measured matched samples, while individual correct answers can still be damaged. On 495 identical, scorable GSM-Plus v3 questions, MiniMax was wrong on **17/495 (3.43%)** initial answers, **9/495 (1.82%)**

after a neutral self-check, and **4/495 (0.81%)** after a separate natural cross-check using DeepSeek’s independently generated reply. Each follow-up branched from the same MiniMax initial conversation. Relative to initial, the paired question-bootstrap differences were **−1.62 percentage points** for self-check (95% interval −3.03 to −0.40) and **−2.63 points** for natural cross-check (−4.24 to −1.21). Self-check corrected/eliminated ten initial errors but introduced two new ones; natural cross-check corrected/eliminated fourteen and introduced one. See the [495-row matched table](#) and [R effect table](#).

On a separate, revised 500-question mathematics bank, MiniMax’s neutral self-review lowered errors from **66/488 (13.52%)** to **42/488 (8.61%)** on the same 488 questions with complete, scorable answers in both rounds. It changed **32 wrong→correct** and **8 correct→wrong**. That bank and its R reproduction are retained privately because they incorporate user-local source material. The original-budget M1 full sample is 70/493 wrong; comparing that directly with M2’s 42/488 would mix denominators. These two banks give evidence of a reduction under their particular designs, not a universal guarantee.

For facts, the newer 500-question collection currently supports **initial-answer** estimates only: DeepSeek **44/500 wrong (8.80%)** and MiniMax **89/496 wrong (17.94%)** in the more comparable completed view; four MiniMax tasks were technically incomplete. A later 500/500 MiniMax supplement is mixed-condition technical recovery. A different, earlier 100-question repeated-model fact study reported self-check **23.25%** versus natural cross-check **5.00%** on 394 matched model-repeat units, with a 97.5% interval for cross minus self of **[−22.00, −14.25] percentage points**. That historical result includes a technical-recovery overlay and is not a full follow-up of the newer 500 facts; see the [separate historical report](#). Neither fact result should be pooled with the mathematical rates above.

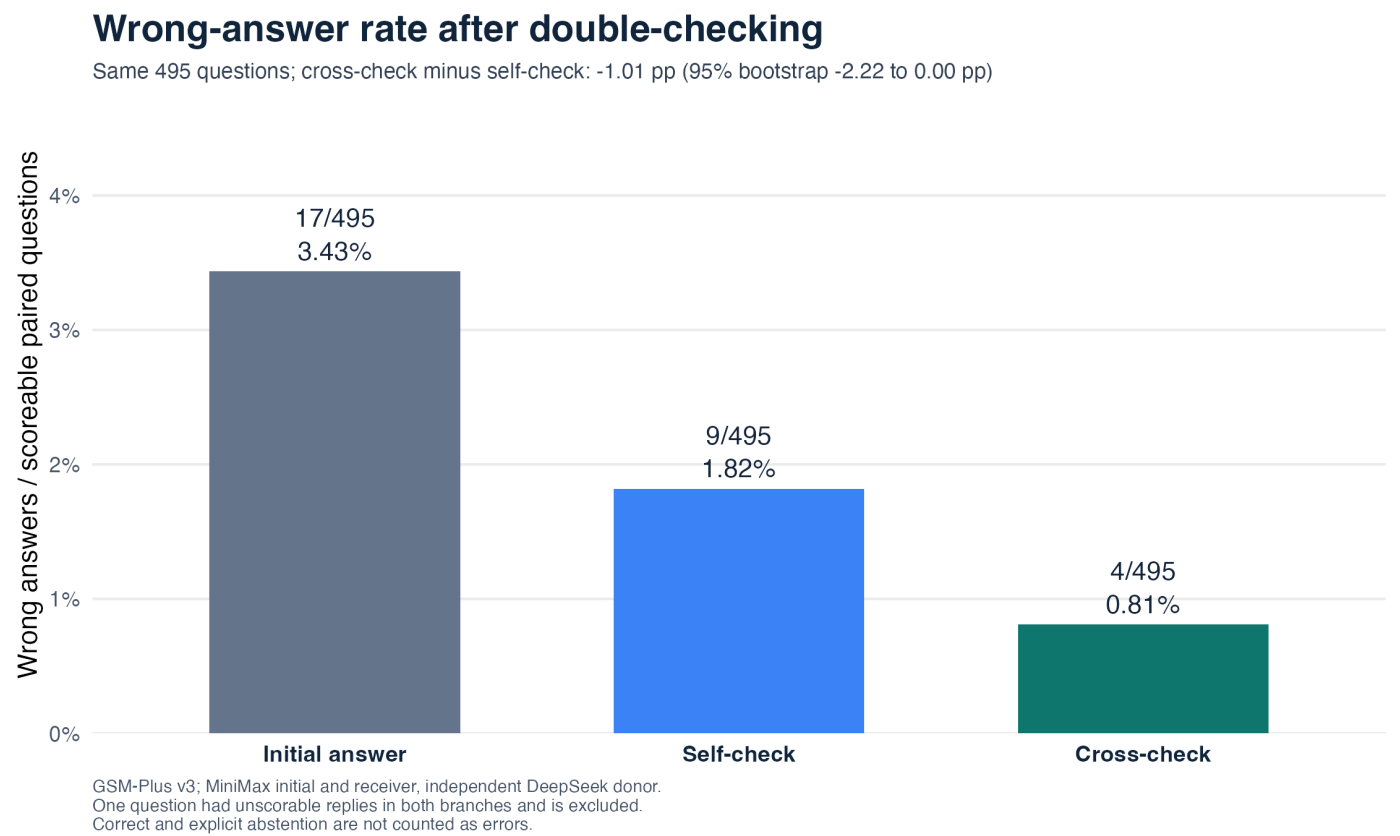


Figure 1. The correct answer and an explicit abstention are both nonwrong; these 495 matched questions have no explicit abstentions.

2. Which double-checking method is most effective in practice?

We cannot establish a universal winner. In the GSM-Plus v3 matched sample, natural cross-check has a lower observed wrong-answer rate than self-check (4/495 vs 9/495). Its paired difference is **−1.01 points**, but the 95% question-bootstrap interval **[−2.22, 0.00]** reaches zero. After a labelled, post-hoc exclusion of four wording-sensitive questions, the difference is −0.61 points with an interval **[−1.63, +0.20]**. The v3 baseline is unusually strong (17/495 wrong), limiting how much either method can improve it. A separate historical fact/math comparison is reported in [its own methods and result summary](#); its receiving models, repeated calls and technical-recovery overlay differ, so it cannot

simply be pooled with this v3 comparison.

In practice, the best choice depends on the question and available evidence. Our experiment compares two **model-only conversational checks**; it does not rank independent source verification, web evidence quality or calculation tools against them. Optional native search was available to each model, but the prompts did not require it. A correct number does not prove that the model actually verified its source or every reasoning step.

In the revised mathematics bank used by the current presentation, natural cross-check corrected **55/63** initially wrong answers versus **31/63** after self-check on the same scoreable questions. That is a targeted correction comparison, not an overall ranking: initially correct answers were not sent through this natural cross-check branch.

3. Can an incorrect peer answer mislead an initially correct model during cross-checking?

It can happen, but the new natural v3 sample did not yield a clean estimate of the risk. In its 495 matched questions, only **two** had a correct MiniMax initial answer and a wrong independent DeepSeek answer. MiniMax rejected one false donor answer. It accepted the other donor's \$900 commission result rather than the frozen \$450 key, but the question's commission-rate wording allows a plausible alternative reading; we do not count that as an unambiguous induced error. The [natural-case review](#) documents both. The older natural cross-check study contains a verified **Polynomial 11** example: DeepSeek first and self-check answered 20, a wrong MiniMax donor answered 24, and DeepSeek changed its cross-check final answer to 24; an independent recurrence calculation supports 20. That is an observed possibility, not an incidence estimate for everyday conversations. See the [historical report](#).

We separately tested **researcher-scripted** false AI advice on 50 v3 questions selected because MiniMax's earlier answer was correct. MiniMax's final answer was wrong on **0/50**, and it adopted the scripted false target on **0/50**. Neutral self-check was also wrong on **0/50** of those same questions. The [50-row reviewed controlled scores](#) describe this selected prompt condition; zero observed events do not establish immunity to wrong peers.

The current revised-bank AI-framed condition used researcher-written false advice on **70 initially correct MiniMax answers**. Its latest approved selection had **0/69 wrong** among scoreable follow-ups and **0 false-target adoptions**. A separate AI-only 70-question extension had **1/70 wrong** from an arithmetic slip, again without adopting the false target. Neither scripted condition is a direct test of a real, independently generated wrong peer answer given to an initially correct MiniMax answer; the current mathematics presentation therefore leaves this RQ open. The historical Polynomial 11 case above is separate evidence of possibility.

4. Can misleading user input cause an otherwise correct model to give a wrong answer?

The current revised-bank presentation tests a **scripted challenge after a correct initial answer**. On 68 jointly scoreable cases, the user-framed challenge produced **10/68 wrong** final answers versus **1/68** after neutral self-check; all ten wrong answers adopted the supplied false target. The user input was written by researchers, not supplied by recruited participants. This post-answer condition differs from the independent first-prompt test below, and its wording also differs from the AI-framed condition.

Yes, we observed individual correct→wrong cases under a scripted false premise. We selected 50 questions on which both models had previously answered correctly, then sent each model two *new, independent first prompts*: the neutral original question, and the same question preceded by a researcher-written false answer and explanation attributed to the user. No previous answer was passed into either fresh conversation. Among the 50 pairs per model, DeepSeek had **0/50** wrong neutral answers and **3/50** wrong false-premise answers (+6 points; one direct adoption of the supplied wrong answer). MiniMax had **1/50** wrong neutral answers and **5/50** wrong false-premise answers (+8 points; four neutral-correct→false-premise-wrong transitions and three direct false-target adoptions). All 100 MiniMax replies were read in full; its raw records and [100-row R adjudication](#) are preserved. The [R bar chart](#) shows only wrong-answer probability with exact numerators/denominators.

These are small, **previously-correct, selected** samples. The two-sided exact paired-discordance p-values are **0.25** for DeepSeek and **0.125** for MiniMax. The MiniMax question-bootstrap interval [2,16] points is descriptive resampling within this sample; it does not override the exact small-discordance result or establish a stable population increase. Three questions (orders 86, 160 and 252) have wording/answer-interpretation concerns. Excluding that **shared post-hoc** set leaves DeepSeek 0/47 vs 1/47 and MiniMax 0/47 vs 4/47; the primary frozen-key rates remain unchanged. See the [R common-sensitivity table](#).

The prompt combines a wrong answer and rationale, so their effects cannot be isolated. Score explanation conflicts separately from final-answer errors; confidence does not prove verification. The older [post-answer challenge study](#) tests a different timing: a false challenge *after* an initially correct answer.

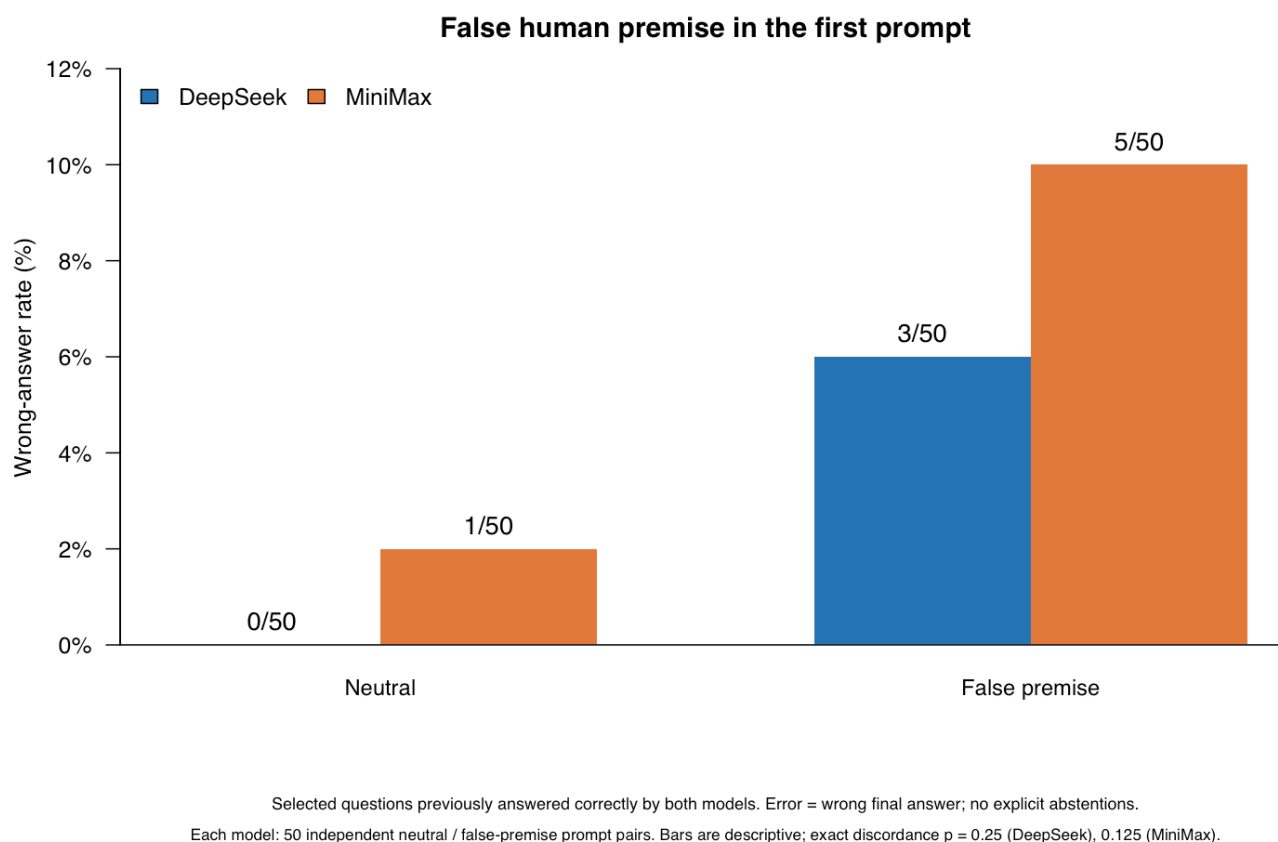


Figure 2. Each model received 50 neutral and 50 false-premise independent first prompts on the same selected questions. The bars show descriptive rates, not a universal effect.

Presentation-safe conclusion

Cross-checking **reduced observed wrong-answer rates** in our matched self/natural studies, but it sometimes introduced new errors and natural cross-check was not shown to outperform self-check reliably across the current v3 comparison. An incorrect peer or user suggestion can redirect a model in particular conversations; in the new 50-question controlled AI-peer v3 subset it did not. Therefore, users should treat AI answers as revisable, check consequential claims against independent evidence when possible, and avoid inserting an unverified answer or explanation into the first prompt. These are recommendations consistent with the observed failure modes, not guarantees that a particular prompting method will make AI correct.

Audit and disclosure

The public v3 self/natural analysis used 255 required full-text decisions among 992 replies (133 screen/key discordances, two nonnumeric, 120 predetermined positive-screen audits). All 120 audited positive screens were correct; the other 737 positive screens remain provisional rather than independently reread. The v3 controlled 50 and MiniMax first-prompt 100 were fully read. The private revised-bank M1–M4 release is reconstructed by R from frozen item annotations, keys and raw replies and matches its historical ledgers row by row; those annotations remain AI-assisted, not an independent human panel. We have not independently proved every benchmark reference key or tested a representative sample of all user prompts. The fact collection’s later MiniMax 500/500 view has **90/500 wrong** but includes three search-disabled technical rescues (one also with a shorter-answer instruction); the more comparable 496-scoreable view is **89/496 wrong**. Both stay versioned in the [inventory](#).